23. Number Sequences & Patterns

Year 4 · Mark Scheme — Supplementary Worksheet

Foundation — Answers

1. 20, 24

2. 60, 70

3. C — 15

4. 15

5. 45, 40

6. 10, 12

7. A, C and E (3,6,9,12 and 9,12,15,18 and 21,24,27,30 all step up by 3; B steps up by 2 and D steps up by 5)

8. 9

9. 60, 50

10. B — add 7

11. 18, 21

12. 30

13. A, C and E (40, 48 and 56 are all multiples of 8 that continue the sequence; B and D are not multiples of 8)

14. 24, 30

15. C — 22

Standard — Answers

16. 48, 96

17. 12, 17 (the gaps increase by 1 each time: +1, +2, +3, so next gaps are +4 and +5)

18. C — double each time

19. 11

20. 20, 10

21. 8

22. A, C and E (5,10,15,20 and 10,15,20,25 and 20,25,30,35 all step up by 5; B and D step up by 4, not 5)

23. 17

24. 55, 45

25. C — 29

26. 60, 72

27. 80

28. 60

29. A — add consecutive odd numbers (the gaps are 3, 5, 7, which are consecutive odd numbers)

30. A, C and E (4, 16 and 36 are square numbers; 10 and 20 are not)

31. 16, 32

32. 20

33. 45, 35

34. B — subtract 9

35. A, C and E (9, 25 and 49 are square numbers; 14 and 30 are not)

Hard — Answers

36. 11 sticks (3 for the first triangle, then +2 for each of the next 4: 3 + 2 + 2 + 2 + 2 = 11)

37. 36, 49 (these are the square numbers 6^2 and 7^2)

38. A and C (3,6,12,24 and 5,10,20,40 both double each time; B, D and E do not)

39. 32 (60, 53, 46, 39, 32 — subtracting 7 four times from the 1st term)

40. 19 edges (4 for the first slab, then +3 for each of the next 5: 4 + 3 + 3 + 3 + 3 + 3 = 19)

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